



Figure 3: Multiplication by AB .

If A is $m \times n$, B is $n \times p$ and x is in $\mathbb{R}^p$, denote the columns of B by $b_1, \dots, b_p$ and the entries in x by $x_1, \dots, x_p$. Then

$$Bx = x_1 b_1 + \dots + x_p b_p$$

By the linearity of multiplication by A ,

$$A(Bx) = A(x_1 b_1) + \dots + A(x_p b_p) \\ = x_1 (A b_1) + \dots + x_p (A b_p)$$

- The vector $A(Bx)$ is a linear combination of the vectors $A b_1, \dots, A b_p$, using the entries in x as weights.
- In matrix notation, this linear combination is written as:

$$A(Bx) = [A b_1 \ A b_2 \ \dots \ A b_p] x$$

Thus multiplication by $[A b_1 \ A b_2 \ \dots \ A b_p]$ transforms x into $A(Bx)$.

Definition: If A is an $m \times n$ matrix, and if B is an $n \times p$ matrix with columns $b_1, \dots, b_p$, then the product AB is the $m \times p$ matrix whose columns are $A b_1, \dots, A b_p$. That is, $AB = A[b_1 \ b_2 \ \dots \ b_p] = [A b_1 \ A b_2 \ \dots \ A b_p]$

This definition makes equation (1) true for all x in $\mathbb{R}^p$. Equation (1) proves that the composite mapping in Figure 3 is a linear transformation and that its standard matrix is AB . Multiplication of matrices corresponds to composition of linear transformations.

Example 3: Compute AB , where $A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}$

Solution: $B = [b_1 \ b_2 \ b_3]$ $AB = [A b_1 \ A b_2 \ A b_3]$ (By Defn.)

$$A b_1 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 4 + 3 \cdot 1 \\ 1 \cdot 4 + (-5) \cdot 1 \end{bmatrix} = \begin{bmatrix} 11 \\ -1 \end{bmatrix}$$

(Row-Vector Rule)